Customer Segmentation Analysis Report

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Tools Used: Python | Scikit-learn | Pandas | Matplotlib
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1. Executive Summary

This project analyzes customer behavior using K-Means clustering to segment customers based on income and spending patterns. The goal is to transform raw customer data into meaningful insights that support targeted marketing and business decision-making.

The analysis identifies distinct customer groups, highlights high-value segments, and uncovers opportunities for improving customer engagement and revenue.

2. Problem Statement

Businesses often struggle to understand diverse customer behavior due to large and unstructured datasets. Without proper segmentation, it becomes difficult to:

- Identify high-value customers
- Target potential customers effectively
- Improve marketing efficiency

This project addresses these challenges using clustering techniques to group customers based on behavior.

3. Objectives

- Segment customers based on income and spending behavior
- Identify high-value and low-value customers
- Detect potential target segments
- Visualize customer groups for better understanding

4. Dataset Overview

The dataset includes:

- Customer ID
- Gender
- Age

- Annual Income (k$)
- Spending Score (1–100)

This enables behavior-based customer segmentation.

5. Tools & Technologies

- Python (Pandas, NumPy)
- Scikit-learn (K-Means Clustering)
- Matplotlib & Seaborn (Visualization)
- CSV File (Dataset)

6. Data Preparation

- Cleaned missing values
- Converted data types where required
- Selected relevant features (Income & Spending Score)
- Prepared dataset for clustering

7. Methodology

◆ **Data Analysis**

- Explored income and spending patterns
- Identified relationships between variables

◆ **Clustering Approach**

- Applied K-Means clustering algorithm
- Used Elbow Method to determine optimal clusters

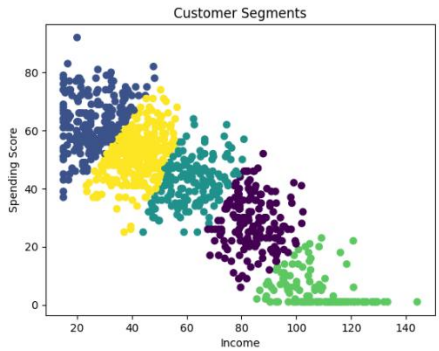
Optimal clusters identified: **K = 5**

8. Model Implementation

K-Means clustering groups customers by minimizing the distance between data points and cluster centers. This allows identification of similar customer behavior patterns.

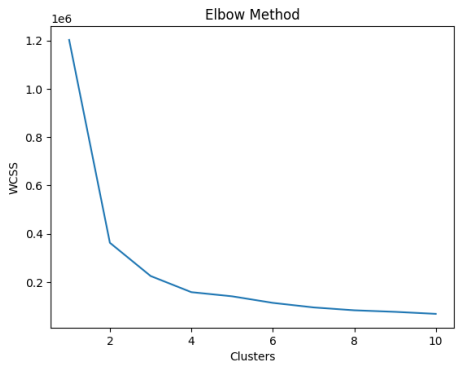
9. Visualization & Analysis

Figure 1: Customer Segmentation (Clusters)



- Shows distinct customer groups
- Each color represents a segment

Figure 2: Elbow Method



- Helps determine optimal number of clusters
- Confirms K = 5

10. Key Insights

- High income + high spending customers are premium and most valuable
- High income + low spending customers are potential targets
- Low income + high spending customers are risky
- Medium income customers show stable behavior
- Customer behavior varies significantly across segments

11. Business Value

This analysis helps businesses:

- Identify high-value customers
- Improve marketing strategies
- Target potential customers effectively
- Increase revenue and customer retention

12. Conclusion

This project demonstrates how machine learning can be used to segment customers effectively. The insights derived can help businesses make data-driven decisions and optimize customer engagement strategies.